

Jacobo Gabeiras

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Nationality: Spanish **Languages:** Spanish (native), English (C1), French (B2)

PROFESSIONAL SUMMARY

Environmental physicist with a PhD from *Université Grenoble Alpes*, specializing in urban climate modelling, Earth Observation (EO) data integration, and open-source scientific software. Developer of *WRFUP*, an open-source Python package that automates EO-based urban parameter generation and ingestion for WRF. My research focuses on developing reproducible tools, datasets, and workflows that bridge Earth observation, urban canopy modelling, and machine learning to improve urban climate simulations.

RESEARCH INTERESTS

Urban climate modelling · Urban data · Earth observation · Scientific software · ML for science

EDUCATION

PhD in Environmental Physics / Climate Modelling, Université Grenoble Alpes, France 2021–2025

– Thesis: *Urban adaptation to extreme heat using WRF-BEP+BEM in Grenoble, France*.

– Supervisors: Chantal Staquet, Charles Chemel.

MSc in Meteorology and Geophysics, Universidad Complutense de Madrid, Spain 2020–2021

– Specialization in atmospheric physics and numerical modelling.

BSc in Physics, Universidad Complutense de Madrid, Spain 2012–2017

– Graduated in the 95th percentile; thesis on mesoscale meteorology.

RESEARCH AND PROFESSIONAL EXPERIENCE

PhD Researcher – Urban Climate Modelling, Université Grenoble Alpes, France 2021–2025

– Designed and developed *WRFUP*, an open-source Python package that automates the ingestion of Earth Observation-based urban parameters into WRF.

– Conducted high-resolution (200 m) WRF simulations to evaluate urban heat mitigation.

– Developed and evaluated urban morphology datasets from EO and LiDAR sources for urban modelling.

– Quantified impacts of adaptation measures on temperature, thermal comfort, and energy use.

– Disseminated results through international conferences (EMS, ICUC) and peer-reviewed journals.

Lecturer – ML for Environmental Fluid Mechanics, Master EFM M2, UGA 2022–2025

– Developed and taught a full Master’s-level course integrating ML with environmental sciences.

– Supervised MSc projects combining ML and environmental modelling applications.

Research Assistant – Urban Climate Modelling, UNSW Sydney, Australia 2024–2025

– Collaborated on urban climate simulations for Australian cities using WRF-Comfort.

– Processed and evaluated Earth Observation and urban morphology datasets for urban modelling workflows.

Intern – GMV Aerospace and Defence S.A.U. Feb–Jun 2021

– Developed aircraft trajectory physics-based forecasting models.

– Applied data-driven prediction methods in aerospace contexts integrating meteorological data.

Technology Risk Consultant – Deloitte Spain 2019–2021

– Applied data management and automation tools to support IT risk analytics for major energy clients.

– Worked within Agile development teams using iterative delivery and version-control workflows.

PEER-REVIEWED PUBLICATIONS

– **Gabeiras, J.**, Staquet, C., Chemel, C., Martilli, A. (2025). *Evaluating urban heat adaptation strategies for extreme heatwaves in complex terrain: A case study of Grenoble, France*. *Urban Climate* (Recently accepted with reference *UCLIM_103059*; preprint available at 10.31223/X5PJ1H).

- Vurro, G., Martilli, A., Hadjinicolaou, P., **Gabeiras, J.**, et al. (2025). *Adaptation and mitigation of outdoor heat stress and building energy consumption during a heat wave in Nicosia, Cyprus*. *Urban Climate*, 58, 102247.

REVIEW PUBLICATIONS

- **Gabeiras, J.**, Staquet, C., Chemel, C., Martilli, A. (2025). *High-resolution simulation of the urban heat island effect in Grenoble during the 2018 heatwave: Evaluating WRF model configurations*. *Boundary-Layer Meteorology* (under review; preprint available at 10.31223/X53N0W).
- **Gabeiras, J.**, Staquet, C., Chemel, C., Martilli, A. (2025). *WRFUP: A Python package to enhance urban simulations*. *Journal of Open Source Software* (under review; preprint available at 10.31223/X57F1F).
- **Gabeiras, J.**, Staquet, C., Chemel, C., Martilli, A. (2025). *A comparative evaluation of advanced urban data methods in WRF*. *Urban Climate* (under review; preprint available at 10.31223/X5F162).

CONFERENCE PRESENTATIONS

- **Gabeiras, J.**, Staquet, C., Chemel, C., Martilli, A. (2025). *WRFUP: WRF Urban Parameters Python Package*. *12th International Conference on Urban Climate (ICUC12)*, Rotterdam, Netherlands, 7–11 July 2025, Oral Presentation ICUC12-1123, DOI: 10.5194/icuc12-1123.
- Fazeli, M., Nazarian, N., Evans, J., **Gabeiras Penas, J.**, Sharma, S., Martilli, A. (2025). *Are High-Resolution Urban Datasets Necessary for Accurate Heat Exposure Modelling in Cities?* *12th International Conference on Urban Climate (ICUC12)*, Rotterdam, Netherlands, 7–11 July 2025, Oral Presentation ICUC12-104, DOI: 10.5194/icuc12-104.
- **Gabeiras, J.**, Staquet, C., Chemel, C., (2024). *Adaptation scenarios to climate change during heatwaves in the Grenoble metropolitan area*. *EMS Annual Meeting 2024*, Abstract EMS2024-1082, DOI: 10.5194/ems2024-1082.

TEACHING AND MENTORING

- Course designer and lecturer, *Machine Learning for Environmental Fluid Mechanics* (Master EFM M2, Université Grenoble Alpes, 2022–2025).
- Supervised MSc theses and internships on urban climate modelling and EO-based data analysis.
- Guest lecturer on urban heat mitigation and adaptation planning (Grenoble Métropole, 2023–2024).

ADDITIONAL TRAINING AND CERTIFICATIONS

- *Machine Learning in Python with scikit-learn* – Inria / FUN-MOOC (2024)
- *Deep Learning with Fidle* – CNRS / Université Paris-Saclay (2023)
- *WRF Tutorial Training* – NCAR Mesoscale & Microscale Meteorology Division (2022)
- *Machine Learning (graduate-level course)* – MSc in Data Science, University of Barcelona (2020)
- *École d'été Autour du 2 °C – Climate and Health Challenges* (2023)
- *Agile Project Management Fundamentals* (2021)
- *MOOC Research Integrity in Scientific Professions* – Univ. Bordeaux (2024–2025)

TECHNICAL SKILLS

Atmospheric Modelling: WRF, BEP+BEM, SLUCM, urban canopy and building energy modelling
Scientific Computing: HPC environments, Git/GitHub, reproducible workflows, open-source software
Programming: Python (xarray, pandas, rasterio, netCDF4, scikit-learn, PyTorch), Bash, Git, LaTeX
Earth Observation & Data: Sentinel-1/2/3, MODIS, WSF3D, LiDAR, PDAL, GDAL, QGIS
Machine Learning: Random Forests, Gradient Boosting, CNNs, feature engineering
Visualization: Matplotlib, Manim, scientific visualization and communication

SELECTED SOFTWARE AND DATA PROJECTS

WRFUP

Open-source Python package that automates the generation and ingestion of Earth Observation-based urban parameters into WRF, supporting reproducible urban climate simulations.

github.com/jacobogabeiraspenas/wrfup

Global Urban Surface ML

Machine-learning workflow for estimating urban morphology parameters from Earth Observation and LiDAR data and generating model-ready urban descriptors.

github.com/jacobogabeiraspenas/global-urban-surface-ml

AWARDS AND SCHOLARSHIPS

- *Excellence Award of Madrid* (2014) – for outstanding academic record.
- *PhD Fellowship* – Université Grenoble Alpes (2019–2025).
- *Doctorate in Environmental Physics* – successfully defended April 2025.

OUTREACH AND COMMUNICATION

- Participant, *Ma thèse en 180 secondes* (2024).
- Presenter at local climate adaptation workshops (Grenoble Métropole, 2023–2024).
- Contributor to open-source climate modelling projects and advocate for reproducible research.

REFERENCES

Prof. Alberto Martilli

Research Scientist
Centro de Investigaciones
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Tecnológicas (CIEMAT)
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